Exercise 5

A walker that walks

At the end, `A` and `D` walk the walker left and right, facing the way it goes, and the walls stop it.

Before you start

- Exercise 4 done and pushed: the walker standing on the floor. In a new session, run `git pull` first

1 Script, run 1: the references

The script grows in three runs; you press Play after each one, so a mistake can only be in the lines you just added.

- Right-click `Assets/Scripts`, **Create** › **Scripting** › **MonoBehaviour Script**, and name it `WalkerController` right away: the class name must match the file name.
- Drag the script onto `Walker`, then double-click it to open it in Visual Studio.

Replace its contents with:

```
using UnityEngine;
using UnityEngine.InputSystem;

public class WalkerController : MonoBehaviour
{
    private Rigidbody2D rb;
    private SpriteRenderer spriteRenderer;
    private InputAction moveAction;

    void Awake()
    {
        rb = GetComponent<Rigidbody2D>();
        spriteRenderer = GetComponent<SpriteRenderer>();
        moveAction = InputSystem.actions.FindAction("Move");
        Debug.Log($"Awake: rb={rb != null}, sr={spriteRenderer != null}, " +
                  $"move={moveAction != null}");
    }
}
```

- Save, go back to Unity and press **Play**: the Console shows `Awake: rb=True, sr=True, move=True`; a `False` points to a missing component or a misspelled action name. Stop.



Run 1: all three lookups found something.

2 Run 2: the input

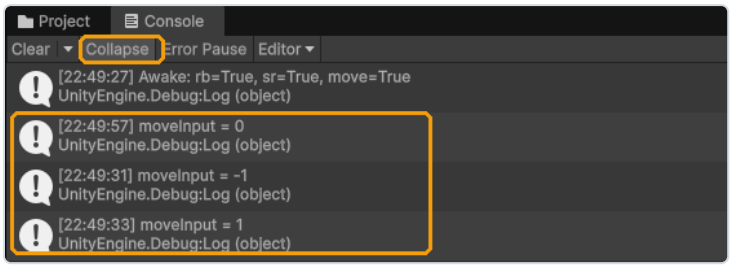
Run 2 only reads the keyboard and prints the value, so you see the input work before anything moves.

Below `Awake`, add:

```
    private float moveInput;

    void Update()
    {
        moveInput = moveAction.ReadValue<Vector2>().x;
        Debug.Log($"moveInput = {moveInput}");
    }
```

- **Play**, click the Game view so it gets the keys, hold `A`, then `D`, then nothing: the Console shows `moveInput = -1`, `1` and `0`. Stop.



With Collapse on, each value is listed once, with its count at the right.

- Delete the `Debug.Log` line from `Update`: it prints every frame.

3 Run 3: motion and facing

Run 3 turns the input into motion: `Update` reads the keys every frame and turns the sprite toward the direction of travel, and `FixedUpdate` moves the body in step with physics.

At the top of the class, add the speed, which then shows in the Inspector:

```
    [Header("Movement")]
    [SerializeField] private float moveSpeed = 6f;
```

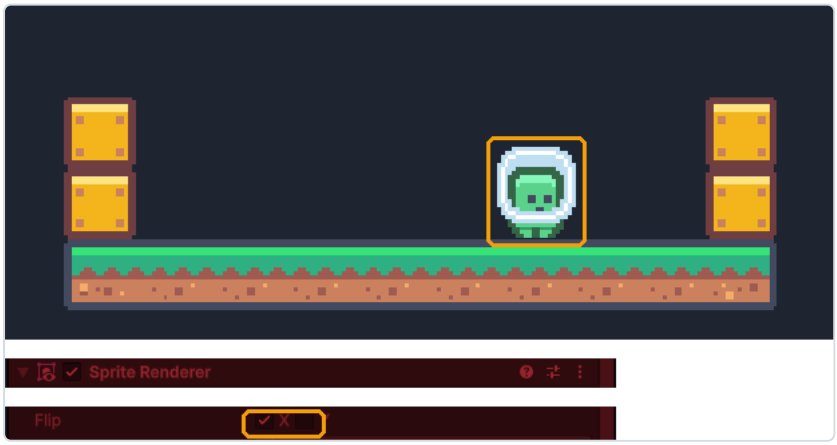
In `Update`, after the `moveInput` line, add the facing; the astronaut faces left in its image, so it is mirrored when it walks right:

```
        if (moveInput != 0f)
        {
            spriteRenderer.flipX = moveInput > 0f;
        }
```

Below `Update`, add (horizontal speed from the input, vertical speed kept):

```
    void FixedUpdate()
    {
        rb.linearVelocity = new Vector2(moveInput * moveSpeed, rb.linearVelocity.y);
    }
```

- **Play** with `Walker` selected and hold `A`, then `D`: the walker slides left and right and stops at the walls. Walking right ticks Sprite Renderer, Flip X, walking left clears it, and letting go keeps the last facing, because the check skips `0`.



Walking right: Flip X is ticked and the astronaut faces right.

- `Ctrl` + `S`, then commit and push with the message `Make the walker walk`.